

Amit Kumar Mohanty

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PROFESSIONAL SUMMARY

AI Engineer with 1+ year of experience designing, building, and deploying intelligent systems powered by LLMs, Agentic AI, and Generative AI. Skilled in Python, LangChain/LangGraph, RAG, fine-tuning (LoRA/QLoRA), and serving AI applications at scale with FastAPI and cloud platforms (AWS, GCP Vertex AI), with a peer-reviewed publication in computer vision. Outside of work, a competitive tennis player and avid sketch artist, pursuits that sharpen discipline, focus, and creativity.

EDUCATION

Birla Institute of Technology and Science, Pilani (BITS Pilani)

July 2026 – July 2028

M.Tech in Artificial Intelligence and Machine Learning (Currently Pursuing)

Relevant Coursework: NLP, Deep Learning, Generative AI, LLMs, Deep Neural Networks, Deep Reinforcement Learning

K.R. Mangalam University

July 2021 – July 2025

B.Tech in Computer Science and Engineering (Specialization in AI/ML), Gurgaon

EXPERIENCE

Research Associate – AI, Keywords Studios India – Gurgaon, Haryana

May 2025 – Present

- Awarded **"Rookie of the Year 2025"** for surpassing **all KPIs** through advanced model training, data-pipeline optimization, and cross-functional collaboration.
- Trained and optimized **enterprise-grade Agentic AI** models to autonomously execute complex multi-step workflows (flight booking, e-commerce ordering, web data retrieval), improving **task-completion reliability**.
- Contributed to a cutting-edge **web-automation AI agent** that visually interprets on-screen content and, from a natural-language instruction, autonomously plans and executes browser actions such as clicking, typing, and navigating to complete complex, multi-step tasks.
- Trained, evaluated, and quality-controlled the agent with a **global team**, curating high-quality datasets and running structured test reviews to deliver client-ready iterations with **improved action accuracy and reliability**.
- Built custom data pipelines and delivered **85%** of self-generated training datasets directly to production as the **first member of a global AI team**, accelerating model readiness.

Artificial Intelligence Intern, Infosys – Remote

November 2024 – February 2025

- Engineered an **AI call-center system** with LangChain and LangSmith, boosting response accuracy by **35%** and elevating customer-AI interaction quality.
- Developed and deployed LLM features that reduced system latency by **25%**, streamlining the AI deployment pipeline.
- Designed and A/B-tested prompt-engineering strategies with automated LangSmith evaluation, reducing **hallucinations by 20%** and making customer conversations feel more natural and trustworthy.
- Collaborated with **12 engineers** to integrate end-to-end AI workflows and was selected to present at the **Best Project by Indian Interns** initiative.

PROJECTS

Dynamic Screen Companion

June 2025 – Present

- Built a **real-time multi-modal AI assistant** integrating the **Gemini API** to analyze live screen activity and deliver continuous voice feedback and interactive support.
- Engineered a high-performance FastAPI model-serving backend handling concurrent inference streams, reducing manual task time by **40%**.
- Designed the end-to-end multi-modal architecture from scratch and optimized OCR pipelines to maintain **95%+ extraction accuracy** across varied layouts.

Diabetic Optiscan

January 2025 – June 2025

- Built an AI tool that scans retinal (eye) images to detect early signs of diabetic retinopathy, and co-authored a **peer-reviewed research paper** explaining how the system was designed and the accuracy improvements it achieved.
- Led integration of a **Vision Transformer (ViT-B/16)** on the APTOS dataset, optimizing self-attention to capture complex spatial patterns in retinal images.
- Designed a custom wavelet modification technique that boosted **detection precision by 50%** over legacy methods.

Currency Detection App for Visually Impaired

July 2024 – December 2024

- Trained an image-classification model on **200,000+ banknote images** achieving **96% accuracy**, deployed in a mobile app with voice feedback to help prevent fraud.
- Conducted **user interviews with visually impaired individuals** to refine real-world accessibility features.

SKILLS

Languages & Frameworks: Python, C++, SQL, FastAPI, TensorFlow, PyTorch, LangChain, LangGraph, Hugging Face, OpenCV, Scikit-learn

Generative & Agentic AI: LLMs, Agentic AI, RAG, Fine-tuning (LoRA, QLoRA), Prompt Engineering, Transformers, BERT, Multi-modal AI, Gemini/OpenAI APIs, NLP, Vector Databases (FAISS/Chroma)

AI Systems & Deployment: Model Serving, API Development, Docker, CI/CD, MLflow, Model Evaluation & Optimization, Data Pipelines, Git

Data & Cloud: AWS, GCP (Vertex AI), MySQL, MongoDB, Pandas, NumPy, Power BI

CERTIFICATIONS

Ducat India – Data Science Professional Training

April 2024 – April 2025

Professional training in Data Science: Python, statistics, machine learning, data analysis, and predictive modeling.

AI/ML for Geodata Analysis – ISRO

Applied ML and deep learning to satellite and remote-sensing imagery for terrain classification and geospatial pattern analysis.